



Koneru Lakshmaiah Education Foundation

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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: DATA ANALYTICS AND VISUALIZATION – 1(24DSB3101)

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TITLE OF THE PROJECT	GARUDA AI — An AI-Driven Crime Pattern Analysis and Public Safety Intelligence System	
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Abstract

Conventional crime prediction systems are largely reactive, often rely on a single data modality, and typically provide risk predictions without sufficient explanation or independent verification. These limitations reduce the transparency, reliability, and practical adoption of AI-based crime intelligence systems. **GARUDA AI: A Multi-Source, Cross-Verified, and Explainable AI Framework for Crime Pattern Analysis and Proactive Public Safety Intelligence** is proposed to address these challenges by integrating multiple sources of crime-related information with predictive analytics, statistical analysis, temporal analysis, spatial analysis, natural language processing, and explainable artificial intelligence within a unified framework. The proposed system aims to support proactive crime pattern analysis and provide decision-ready public-safety intelligence.

The framework will integrate **structured crime records, geospatial data, weather and census information, and textual data from news and social media** through a unified data pipeline. The proposed ETL process will include data cleaning, missing-value handling, categorical encoding, geocoding, normalization, duplicate detection, outlier handling, and feature preparation. Exploratory and statistical analysis will be performed to identify crime distributions, trends, correlations, relationships, and significant risk-associated factors. The textual data component will also be used to identify relevant crime and threat signals from news and social-media content, providing a broader information base than systems that depend only on structured crime records. The proposed framework will utilize benchmark datasets including **UCI Communities and Crime, Chicago Crime (2001–Present), and NCRB Crime in India**.

For crime-risk estimation, GARUDA AI will employ a **multi-model predictive core** consisting of **Logistic Regression, Decision Tree, Random Forest, XGBoost, LightGBM, and CatBoost**. Instead of accepting the output of a single classifier as the final prediction, the framework will introduce a **Cross-Verification Consensus Layer**.

High-risk predictions will be independently examined from statistical, temporal, and spatial perspectives using **Isolation Forest for anomaly verification, LSTM for temporal verification, and DBSCAN and K-Means for spatial hotspot verification**. The proposed approach will use agreement among these independent analytical perspectives to strengthen the reliability of flagged predictions. Model performance will be evaluated using **Accuracy, Precision, Recall, F1-Score, ROC-AUC, Confusion Matrix, cross-validation, and cross-model agreement**, providing a broader assessment of predictive performance and reliability.

To ensure that predictions are transparent, interpretable, and auditable, GARUDA AI will incorporate a comprehensive **Explainable AI layer** using **SHAP, LIME, Feature Importance, Partial Dependence, and Counterfactual/Anchor explanations**. These techniques will help identify the factors influencing predictions, explain individual risk assessments, analyze feature-risk relationships, and demonstrate how changes in relevant input factors may influence predicted outcomes. The resulting intelligence will be presented through an **interactive dashboard** containing crime-risk information, hotspot maps, temporal trends, analytical visualizations, model explanations, and reports. A proposed multi-channel alert mechanism using **email, SMS, and push notifications** will also be incorporated to communicate relevant verified intelligence to authorized users.

The primary innovation of GARUDA AI is the integration of **multi-source data fusion, multi-model prediction, cross-verification, and multi-technique explainability within a single analytical pipeline**. The proposed Cross-Verification Consensus Layer is intended to ensure that a high-risk prediction is not surfaced solely on the basis of one classifier, but is examined using statistical, temporal, and spatial evidence before being presented to a decision-maker. The framework aims to progress beyond static crime prediction by providing transparent, cross-verified, and decision-ready intelligence for **crime-risk assessment, hotspot identification, emerging pattern analysis, proactive public-safety planning, and informed resource allocation**. Through this integrated approach, GARUDA AI is proposed as a responsible and interpretable AI-based framework that can support authorized law-enforcement and public-safety decision-makers while ensuring that AI-generated predictions are treated as explainable and verifiable intelligence rather than unquestionable decisions.

Signature of the Guide

HOD